

KDD: A kernel density based descriptor for 3D point clouds

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ABSTRACT

3D feature description is one of the central techniques that rely on point clouds since a lot of point cloud processing techniques apply the point-to-point correspondences that are achieved via feature descriptors as input data. The feature descriptor encodes the information of the underlying surface around the feature point so as to make a local surface distinguished from another. The focus of the existing descriptors is accumulating the geometric or topological measurements into histograms or encoding the 2D images that are acquired by rotationally projecting the 3D local surfaces onto 2D planes. Histograms can hardly deal with three or more dimensional information, and the rotational projection operation does bring much unnecessary intermediate computations. To overcome these limitations, in this article, a descriptor named Kernel Density Descriptor (KDD) has been presented. One core contribution of this method is to encode the information of the whole 3D space around the feature point via kernel density estimation, and another is providing the strategy for selecting different matching metrics for datasets with diverse levels of resolution qualities. We compare KDD against several representative descriptors on publicly available datasets, the experimental results demonstrate that the KDD descriptor achieves a satisfactory and balanced performance in terms of descriptiveness, robustness, and compactness, furthermore, the comparisons validate the overall superiority of our method. The benefits and applicability on object registration and recognition and 3D object reconstruction are demonstrated by the favorable results that are obtained for both public datasets and the real-world point clouds of Terracotta fragments.

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1. Introduction

Feature point descriptor for point clouds is a fundamental representation structure in 3D modeling [1], 3D object recognition [2], and 3D model registration [3]. Therefore, the 3D local feature descriptors construction is at the core of many computer graphics and computer vision technologies.

To be useful for downstream tasks, the descriptors need to satisfy several performance requirements, such as descriptiveness, robustness, compactness, efficiency and the memory footprint [4]. Among these requirements, descriptiveness and robustness are considered to be two of the most essential aspects of a descriptor. A feature descriptor is descriptive if it can encode predominant and sufficient information about the local surface. That is to say, a descriptor should provide sufficient information to make a local surface distinguished from another. A feature is robust if it is in-

sensitive to several factors that can disturb the surface, e.g., noises and non-uniformly sampling density.

Local features have been extensively investigated over the past few decades to design descriptors that are distinctive and robust, thus the existing descriptors have been comprehensively evaluated and analyzed to improve all aspects of performance [4]. So far, there are mainly three modes to directly construct 3D local feature descriptors, except for deep learning-based methods.

One best-known group of descriptors is the histogram-based descriptors, for example, Signature of Histogram of Orientations (SHOT) [5]. Histogram is a very simple and functional representation for one or two-dimensional information, thus making it practical for the density estimation and rapid visualization of data in one or two dimensions. However, the drawbacks of histograms are a bit obvious: (1) The starting location and bandwidth of the bins seriously affect the performance; (2) the curse of dimensionality leads to the number of bins grows exponentially with the number of dimensions; and (3) histograms can hardly deal with three or more dimensional information, which is a much more serious problem in representation tasks in the 3D domains.

Another category of descriptors employs binary patterns to describe the underlying surfaces, including 3D Local Binary Patterns

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(3D-LBP) [6] and Binary SHOT (B-SHOT) [7], et al. Binary descriptors prevail over histogram-based descriptors in terms of compactness and efficiency, but achieve relatively lower scores in descriptiveness [8]. Furthermore, several 3D binary descriptors apply to the voxel model only [6].

Alternatively, the rotation and projection mechanism generates descriptors by first repeatedly rotating and projecting local surface onto 2D planes and then encoding the producing 2D images or range images. A representative descriptor is Rotational Projection Statistics (RoPS) [9]. The rotation and projection mechanism also gets satisfactory results, however, the rotation and projection operations do bring lots of unnecessary intermediate computations.

From an intuitive point of view, the geometry of the underlying surface around a feature point is shaped by the distribution of its neighboring points. That is to say, different shapes have different point's densities in different regions of the 3D space around the feature points. Furthermore, two points with the same Euclidean distance may form different shapes. This is because the points' contribution to the underlying shape differs on different axes of the local reference frame (LRF). Based on the observations above, instead of computing the geometric attributes of points in the support region, we convert the description of the feature point to the kernel density estimation of the 3D space around the feature point, with different point's densities computed in different regions. Therefore, a novel descriptor that is termed kernel density descriptor (KDD) is proposed in this work. First, we compute the LRF at the feature point and exploit a cubic support region around the feature point that is aligned with the LRF. Then the kernel density of each cube in the 3D space is estimated via kernel density estimation which overcomes the limitations of histograms. Note that, in order to promote the robustness to point non-uniformity, instead of only considering the non-empty cubes as in [10], all the cubes in the support region participate in the density estimation, no matter it is empty or not. Moreover, rather than evaluating different feature descriptors using different matching metrics [11], the KDD was combined with different matching metrics for different datasets and the strategy for selecting different matching metrics for datasets with diverse levels of resolution qualities was provided. We compare the proposed KDD against several representative descriptors on publicly available datasets, and the experimental results demonstrate that the KDD descriptor is more descriptive and achieves the highest recalls compared to the tested descriptors on the selected datasets, for both at different levels of noises and at different simplification rates. Besides, the KDD provides invariance to rigid transformations and achieves satisfactory results in terms of robustness and compactness. We also apply the proposed method on a real-world dataset, i.e. the Terracotta fragment models, and the favorable results demonstrate the effectiveness of KDD and highlight the utility of the proposed method.

The main contributions of this work include:

- (1) the presentation of a 3D local descriptor (KDD) which achieves a satisfactory and balanced performance in terms of descriptiveness, robustness, and compactness, furthermore, it is simple since it encodes the spatial distribution of points, avoiding computing any geometric attributes and needing no rotational projection operations;
- (2) the presentation of the strategy for selecting different matching metrics for datasets with diverse levels of resolution qualities. In particular, difference-based, correlation-based, and entropy-based distances are all suitable for datasets with a high or medium level of resolution quality. However, only correlation-based (such as Cosin distance and Pearson correlation coefficient) and entropy-based distances (for example, KL divergence) are suitable for datasets with a low level of resolution quality.

2. Related works

Several researchers have devoted great efforts to solve the local surface description problem in the past few decades and a diverse of 3D local descriptors are presented. Among the state-of-the-art descriptors, some are generated directly on the point clouds by encoding the information of the local surfaces into histograms, whereas the others are constructed on 2D images acquired by rotating and projecting the underlying surfaces onto 2D planes. Therefore we categorize these algorithms into α descriptors generated on point clouds and β descriptors generated on projected 2D images. Moreover, Deep learning(DL) methods have achieved remarkable performance on the 3D data, which are classified into the third category. This section briefly reviews the existing representative 3D local feature descriptors of the above categories.

2.1. Descriptors generated on point clouds

The early studies generally focus on the meshes that are reconstructed from point clouds, for example, 3D tensor descriptor [12], whereas the recent methods directly deal with the point clouds. We classify the existing approaches into three main categories: histogram based descriptors, binary descriptors and vector-field based descriptors.

2.1.1. Histogram based descriptors

These descriptors describe the underlying surfaces by constructing histograms according to the spatial distributions of data points around the feature points. They usually start with a Local Reference Frame/Axis (LRF/A) generated on the feature point, then divide the 3D support regions into several bins along the three dimensions of the LRF/A, and finally accumulate the spatial distribution measurements of the points (e.g. point numbers or weighted point numbers) on the local surfaces into the histograms.

Spin Image (SI) [13] utilizes the normal vector $\vec{n}_p$ of the feature point p as the LRA, and then accumulates the neighboring points q_i into each bin of the 2D array that is formed according to the in-plane and out-plane distances of the points in the support region. Tri-Spin-Image (TriSI) [14] adopts the procedure for generating SI descriptor but respectively generates three spin images using three axes of LRF. Then the TriSI descriptor is formed by concatenating the three spin images, which significantly improves the descriptiveness and robustness. 3D Shape Context (3DSC) [15] also uses the normal vector $\vec{n}_p$ of the feature point p as its LRA. The generation of the 3DSC contains three phases: first, a 3D spherical grid centered at p , with the north pole of the grid being aligned with $\vec{n}_p$ is created; the support region is then divided into several bins along the radial, azimuth and elevation dimensions respectively; and the weighted number of points falling into each bin of the 3D grid is finally counted. Unique Shape Context(USC) [16] uses an analogous approach to that used in 3DSC, however, the main difference is that USC improves memory footprint and efficiency.

The above-mentioned descriptors deal with the spatial distribution information of points in the support regions, whereas another group of descriptors exploits the geometric attributes of the local surfaces. The main difference among the geometric attribute histogram-based descriptors is the geometric quantities that are used.

Local Surface Patch (LSP) [17,18] accumulates two different information into particular bins along the two dimensions, one is the shape index of point q_i in the support region, and another is the cosine of the angle between the normal of q_i and the normal of feature point p . Thus LSP is a 2D histogram. Thrift [19] is a 1D histogram of the deviation angles between the surface normal at the feature point p and the surface normals of its neighboring points q_i . PFH [20] calculates four geometric quantities for each point pair

in the support region and accumulates points in particular bins along the four dimensions. In a later work [21], Fast Point Feature Histogram (FPFH) which is similar to PFH, is presented. However, instead of considering all point pairs in the support region, FPFH only considers the relationships between the feature point and its neighbors. SHOT [5] first divides the support region into several volumes along the radial, azimuth and elevation axes, and then counts the point numbers according to the angles between the normals of the neighboring points within the particular volume and the normal at the feature point. Finally, all the local histograms form the SHOT descriptor.

The histogram-based descriptors are computational complex and memory consuming, however, they achieve better performance in terms of descriptiveness and robustness compared to the binary descriptors [22].

2.1.2. Binary descriptors

Due to the non-uniformly sampling of the point clouds, designing and applying a fixed pattern on a selected sampling pair are not easy tasks. Matsuda [6] first extends and adapts the binary descriptors BRIEF, ORB and LBP to 3D voxel models, resulting in corresponding 3D-BRIEF, 3D-ORB, and 3D-LBP. LoVS is also a local voxelized structure for 3D binary feature representation [23], which is proved to be distinctive and robust to noises, mesh resolution variation and point irregularity. However, the voxelization operation is very time-consuming.

To convert the floating vector into binary patterns, Manoj et al. apply a binary quantization method on SHOT, RoPS and FPFH. Thus the B-SHOT [7], B-RoPS, and B-FPFH [24] are built. The converting operation does lose information, resulting in the decline of the descriptiveness and robustness.

Strivastava [25] projects the normals of local ordered neighboring points on the LFA of the feature point, whose relative magnitudes are employed to assign binary digits. The generated descriptor is redundant and sensitive to noises.

Binary descriptors prevail over the histogram-based descriptors in terms of compactness and computational-efficiency, however, losing information or producing redundant information can not be avoided.

2.1.3. Vector-field based descriptors

Vector-field based descriptors are often formed via encoding differential invariants. For example, 3D-Div is constructed as the divergence of the reoriented 3D vector field that is a normalized 3D vector field computed at each point in the support region [26], and 3D-Vor exploits the vorticity of the vector field computed at each point to encode the distinctive features at each point on the local surface [27].

2.2. Descriptors generated on projected 2D images

By projecting the local surfaces onto 2D planes, we can obtain 2D images or 2D depth images. It is easier to process the 2D images or 2D depth images than the 3D local surfaces. This idea has been adopted by several researchers to solve the local features representation problem.

RoPS [9] employs the rotation and projection mechanism to encode the local surfaces. Specifically, the local surface is rotated around the three axes of the LRF and then projected onto the three coordinate planes of LRF. For each rotation and projection, a distribution matrix is generated by dividing the corresponding plane into several grids and counting the point number of each grid. Finally, five statistics of each distribution matrix are computed and all these statistics of all rotations and projections are aggregated into a feature vector and thereby forming the RoPS descriptor.

The rotation and projection mechanism highly improves the performance, however, projection from 3D to 2D does lead to geometric information loss. By integrating the depth information of 2D images, Triple Orthogonal Local Depth Images (TOLDI) [28] which is a 1D histogram is built by concatenating local depth images captured from three orthogonal view planes in the LRF. TOLDI is highly robust and efficient, but the dimension can be up to 1200.

Also adopting the rotation and projection mechanism, Binary Rotational Projection Histogram (BRoPH) [8] integrates the density distribution and depth information. Further, an improved LBP is presented to generate binary strings. The advantage of BRoPH is the compactness of representation and efficiency of computation.

To some extent, the rotation and projection mechanism, along with the integration of density distribution and depth information, can improve the descriptiveness and robustness. However, rotation and projection operations do bring much intermediate computation, and the rotation angles affect both the descriptiveness and computational-efficiency.

2.3. Deep learning methods

The focuses of DL methods are mostly the object recognition, classification and segmentation, such as 3D ShapeNets [29] and VoxNet [30], et al. For the 3D feature description, a common strategy is to extract low-level descriptors and then, use them as input to a DNN in order to get a more effective high-level representation for recognition, retrieval, or other tasks, for example, the work described in [31]. Another effective strategy is to directly exploit the 3D models. For example, Convolutional AutoEncoder Extreme Learning Machine (CAE-ELM) combines the strengths of CNNs, AEs, and ELMs to learn a 3D descriptor. PerfectMatch [32] and D3Feat [33] utilize the datasets with labeled tags to train the network, and finally the correspondence search can be achieved. However, 3D models with labeled tags are commonly used in machine learning methods but the creation of the labeled tags is time-consuming, making it difficult to use for some practical applications. The reader is referred to [34] for more details about the DL methods that are applied on 3D models.

3. Methodology

3.1. Overview

The unorganized point cloud is the processing target throughout the life cycle of the proposed method (as shown in Fig. 1(a)), which means that the proposed method deals with no other data types such as 2D images, range images or meshes. The construction of the proposed KDD is very simple since it only contains two phases (please refer to Fig. 1 for the pipeline), without any geometric attributes estimation and rotational projection operations.

First, the proposed method computes the LRFs at key points, and then divides the support region into a series of cubes, which is termed cubic support region in this work (see Fig. 1(b)). The cubic support region serves to the density estimation on the one hand and enhances the robustness on the other.

Second, the kernel densities of the local cubes are estimated (Fig. 1(c)), which are then aggregated into a vector; after that, the float-valued vector forms the KDD descriptor (see Fig. 1(d)). In the kernel density estimation step, in order to cope with the point non-uniformity which occurs quite often with point clouds, the point's densities of some selected spatial points are first computed, which are then employed to estimate the kernel densities of the local cubes. Moreover, to further enhance the robustness to noise and non-uniform sampling, the Gaussian function is utilized as the kernel function.

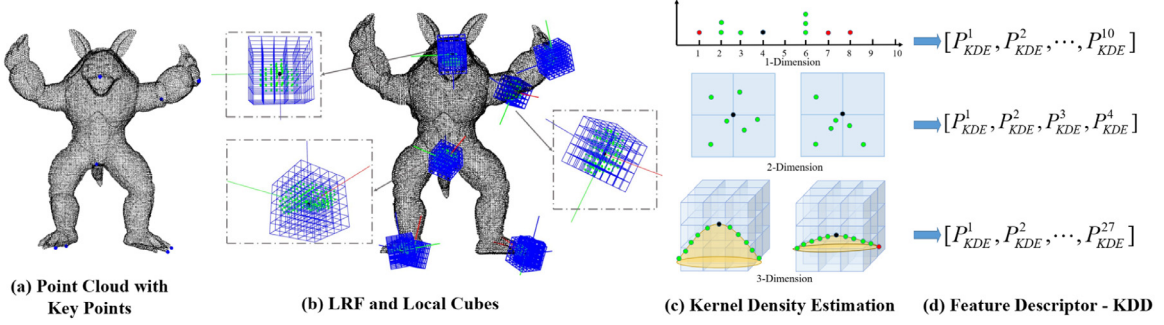


Fig. 1. The pipeline of the proposed method. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

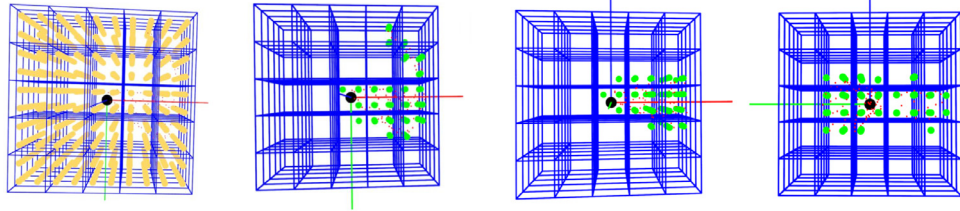


Fig. 2. An illustration of cubic support region and point's density computation. The black ball is a key point. The first figure on the left shows the selected spatial points that are marked in yellow, and the second to the forth figures are three orthogonal views. The blue, red and green lines from the key point are the z-axis, the x-axis and y-axis respectively. The red little points are points in the support region, and the green balls represent the selected spatial points having non-zero point's densities. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

3.2. LRF generation

The generation of the LRF is the first concern for even all of the descriptors. The stability and robustness of the LRF significantly affect the performance of the descriptors. Recently, an LRF is often constructed either based on covariance analysis [8,16,28], or based on the normal vector of the key points [15,23].

The recent descriptor BRoPH [8] constructs the LRF through weighted covariance analysis of points in the support region, replacing the centroid with the key point and assigning nearby points larger weights. The eigenvector $\vec{n}_p$ corresponding to minimal eigenvalue of the covariance matrix is employed as the z-axis. The remaining x-axis and y-axis are constructed through a weighted sum of the projection of the local surface.

We also adopt covariance analysis technique for the generation of LRF. However, integrating the weights that are computed according to the distances between points may result in unstable LRFs on the dataset with a low resolution quality. This is because the local imbalances in the sampling pattern strongly affect the distances between points in the support region, which lead to different spatial distribution information of the same underlying surfaces and therefore, resulting in unstable LRFs. Hence, we construct LRF as follows.

Given a key point p and a support radius r , the points p_i in the ball-neighborhood within r form the local surface and support region of p . Then, the key point p is utilized as the centroid and covariance analysis is performed on the points in the support region. Therefore, the covariance matrix $\mathbf{M}$ of p is computed as:

$$\mathbf{M} = \frac{1}{N-1} \sum_{d_{pp_i} < r} (p_i - p)(p_i - p)^T \quad (1)$$

where N is the number of points in the support region, and d_{pp_i} is the Euclidean distance between points p_i and p .

Then the z-axis of LRF is computed as:

$$\vec{z}_p = \begin{cases} \vec{n}_p, & \text{if } \vec{n}_p \cdot \sum_{i=1}^N p_i \vec{p} \geq 0 \\ -\vec{n}_p, & \text{otherwise} \end{cases} \quad (2)$$

where $\vec{n}_p$ is the eigenvector corresponding to minimal eigenvalue of $\mathbf{M}$, $p_i \vec{p}$ represents the vector between points p_i and p .

The x-axis $\vec{x}_p$ is the sum of the $p_i \vec{p}$, which is computed as:

$$\vec{x}_p = \sum_{i=1}^N p_i \vec{p} \quad (3)$$

finally, the y-axis is calculated via the cross-product between x-axis and z-axis.

3.3. Local cubes generation

Given the LRF, the local cubes (LCs) are generated along the three axes. The LCs play a significant role in the computation of the KDD descriptor for a few reasons: (1) they are used as the Parzen window that is a necessary element for kernel density estimation; and (2) the number of the LCs determines the bandwidth of Parzen window and the dimension of the KDD descriptor.

The local cubes generation requires two sweeps.

In the first sweep, an origin cube C of size $2r \times 2r \times 2r$ is centered at the key point. The sides of the origin cube are parallel to the LRF that is constructed in the previous step. The side length of the original cube is set as $2r$ in order to cover the whole support region.

In the second sweep, the original cube of size $2r \times 2r \times 2r$ is divided into $CubeN^3$ cubes, denoted by c_1^i , as shown in Fig. 2. Note that all of the neighboring points in the support region are in the cubes, since side length of the original cube is set as $2r$ and all of the neighboring points p_i have $d_{pp_i} < r$.

3.4. Kernel density estimation

The core idea of KDD is to estimate the kernel density of each cube c_1^i with side length $l_1 = \frac{2r}{CubeN}$ that is generated in the previous step. In order to capture more detailed information of the underlying surface without increasing the dimension of the KDD (the number of the cubes c_1^i), and cope with the point non-uniformity which occurs quite often with point clouds, we first compute the

point's density of each cube c_1^i instead of directly counting those points. Then the kernel densities of all the cubes c_1^i are estimated for describing the shape of the underlying surface around the key point.

The point's density is computed as follows: first, we further divide each cube c_1^i into n^3 cubes (denoted by c_2^j) and we set $n = 2$ in the experiments presented in this work. Centre points (denoted by cp_2^j) of the cubes c_2^j with side length $l_2 = \frac{2r}{n \times \text{CubeN}}$ after second division are then computed, which are employed as the selected spatial points, as shown in Fig. 2. After that, the point's density $D_{cp_2^j}$ of each selected spatial point cp_2^j is calculated as:

$$D_{cp_2^j} = \frac{\#Num_{points}}{\#Num_{ave}} \quad (4)$$

where $\#Num_{points}$ represents the number of neighboring points within $1.3 \times l_2$ of the selected spatial points cp_2^j , and $\#Num_{ave}$ is the average point's density that is the average number of points within d_p on the given point cloud. d_p is the average sampling that is computed as:

$$d_p = \frac{1}{|N_p|} \sum_{p_j \in N_p} |p - p_j| \quad (5)$$

N_p is the k nearest neighbors of point p and we set $k = 20$ in this work. At last, the point's density $D_{c_1^i}$ of cube c_1^i is the sum of $D_{cp_2^j}$ of cp_2^j in the cube c_1^i .

Given the point's density $D_{c_1^i}$ of each cube c_1^i , the kernel density $P_{KDE}(c_1^i)$ of each cube c_1^i is estimated using Eq. (6):

$$P_{KDE}(c_1^i) = \frac{1}{\text{CubeN}^3 * n^3} \sum_{j=1}^{\text{CubeN}^3} D_{c_1^i} * K(|c_1^i - c_1^j|) \quad (6)$$

where $K(\mu)$ is the kernel function, and $|c_1^i - c_1^j|$ is defined as the Euclidean distance between the cubes c_1^i and c_1^j . Gaussian function defined in Eq. (7) is used as the kernel function since it is a smooth function that has several advantages: (1) it yields density estimates that have no discontinuities; (2) it unequally weights the cubes c_j according to their distance to c^i ; and (3) it is robust to noise.

$$K(\mu) = \frac{1}{(2\pi)^{D/2}} \exp(-\frac{1}{2}\mu^2) \quad (7)$$

A total of CubeN^3 kernel density estimates of cubes c_1^i ($1 \leq i \leq \text{CubeN}^3$) are aggregated into a vector and thereby forming our descriptor KDD for the key point, as shown in Fig. 1(d). KDD is thus a $1 \times \text{CubeN}^3$ float-valued vector.

3.5. The similarity measurement of KDD

Two points are validated as matched correspondences only if the distance between their feature descriptors is small enough. The KDD is a float-valued vector, thus a variety of simple and typical matching metrics can be utilized.

As indicated in [11], different descriptors with different matching metrics result in different matching performances. However, the performance of the same descriptors with the same matching metrics on different point clouds with diverse levels of resolution qualities is also different. Under this consideration, five typical matching metrics are provided to perform the similarity measurement of KDD: the Euclidean distance (ED), KL divergence (KL), Cosin distance (Cos), Chebyshev distance (Cheby), and Pearson correlation (Pearson). The performances of these metrics on different datasets are tested in Section 4 to determine which metrics should be selected for different datasets. These metrics are respectively defined as:

$$ED = \sqrt{\sum_{i=1}^{\text{CubeN}^3} (P_{KDE}^{p \in C}(c_1^i) - P_{KDE}^{q \in Q}(c_1^i))^2} \quad (8)$$

$$KL = \frac{D_{KL}(p||q) + D_{KL}(q||p)}{2} \quad (9)$$

The $D_{KL}(p||q)$ and $D_{KL}(q||p)$ can be computed as follows:

$$D_{KL}(p||q) = \sum_{i=1}^{\text{CubeN}^3} P_{KDE}^{p \in C}(c_1^i) * \log \frac{P_{KDE}^{p \in C}(c_1^i)}{P_{KDE}^{q \in Q}(c_1^i)} \quad (10)$$

$$D_{KL}(q||p) = \sum_{i=1}^{\text{CubeN}^3} P_{KDE}^{q \in Q}(c_1^i) * \log \frac{P_{KDE}^{q \in Q}(c_1^i)}{P_{KDE}^{p \in C}(c_1^i)} \quad (11)$$

$$\text{Cos} = 1 - \frac{\sum_{i=1}^{\text{CubeN}^3} (P_{KDE}^{p \in C}(c_1^i) * P_{KDE}^{q \in Q}(c_1^i))}{\sum_{i=1}^{\text{CubeN}^3} (P_{KDE}^{p \in C}(c_1^i)^2) + \sum_{i=1}^{\text{CubeN}^3} (P_{KDE}^{q \in Q}(c_1^i)^2)} \quad (12)$$

$$\text{Cheby} = \max_i (|P_{KDE}^{p \in C}(c_1^i) - P_{KDE}^{q \in Q}(c_1^i)|) \quad (13)$$

$$\text{Pearson} = 1 - \frac{E((P_{KDE}^{p \in C} - \mu_{P_{KDE}^{p \in C}}) * (P_{KDE}^{q \in Q} - \mu_{P_{KDE}^{q \in Q}}))}{\sigma_{P_{KDE}^{p \in C}} * \sigma_{P_{KDE}^{q \in Q}}} \quad (14)$$

where point p is on the point cloud C and point q is on the point cloud Q . μ and σ are the mean value and standard deviation.

4. Experimental results

This section is organized as follows. The experimental setup is first introduced in detail, including datasets, descriptors, evaluation criteria used in the experiments and the experimental procedures. Second, the performance of KDD is evaluated and compared with several representative descriptors, in terms of the descriptiveness, compactness, robustness and time efficiency. Finally, we summarize the overall performance of the proposed KDD.

4.1. Experimental setup

4.1.1. Datasets and descriptors

We perform the experiments on three publicly available datasets with different features, scales, sampling qualities, and acquisition ways in order to make comprehensive evaluations and comparisons. The datasets utilized in the experiments include Bologna object recognition dataset, SpaceTime dataset and Kinect dataset. The Bologna dataset is acquired by synthesis thus with a high sampling quality, whereas the SpaceTime dataset and Kinect dataset are captured by a SpaceTime Stereo scanner and a low-cost Microsoft Kinect scanner respectively, they therefore contain images with a medium level and a low level qualities. The detailed information about the datasets is listed in Table 1, and some of the models taken from these datasets are presented in Fig. 3.

For the comparison of performance, we select the typical SHOT and FPFH, which are histogram-based descriptors, as the baseline. Besides, since the KDD describes the local surfaces directly without no projection and rotation operations, therefore, the comparisons between the RoPS and BRoPH, which generate the descriptors by projecting the local surfaces onto 2D planes, are also conducted. Furthermore, SHOT, FPFH, RoPS and KDD are all float-valued vectors, whereas BRoPH is a binary descriptor, the compactness thereby could be evaluated and compared in-depth.

The proposed KDDs with different parameter settings (denoted by KDD[CubeN]) participate in the comparisons. Furthermore, the performance of KDDs with different matching metrics are also tested on all the three datasets. For the sake of clarity, the KDDs denote KDD[5]_KL, KDD[5]_ED, KDD[5]_Cos, KDD[5]_Cheby, and

Table 1
Datasets used in the experiments.

Data Sets	Quality	Acquisition	Models	Scenes	#Models	#Scenes
Bologna	High	Synthesis	3D	3D	6	45
Space Time	Medium	SpaceTime Stereo	2.5D	2.5D	6	12
Kinect	Low	Microsoft Kinect	2.5D	2.5D	6	17

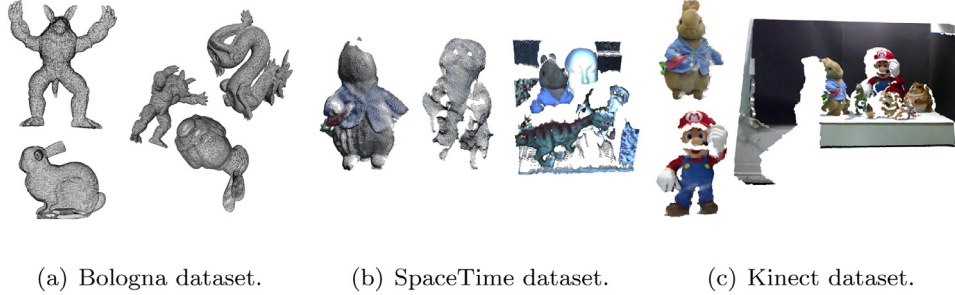


Fig. 3. The datasets used in the experiments. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 2
The selected descriptors.

No.	Descriptors	Classification	Type	Dimension
1	FPFH	Histogram based	float	33
2	SHOT	Histogram based	float	352
3	RoPS	Rotation & Projection	float	135
4	BRoPH	Rotation & Projection	binary	432
5	KDD[4]	Kernel density based	float	64
6	KDD[5]_KL	Kernel density based	float	125
7	KDD[5]_ED	Kernel density based	float	125
8	KDD[5]_Cos	Kernel density based	float	125
9	KDD[5]_Cheby	Kernel density based	float	125
10	KDD[5]_Pearson	Kernel density based	float	125
11	KDD[6]	Kernel density based	float	216

KDD[5]_Pearson respectively. KDD[4] and KDD[6] are matched by using KL divergence, moreover, KDD[5]_KL is utilized for the Robustness test since KL divergence achieves balanced performance on the selected datasets.

All the selected descriptors are implemented in PCL Version 1.8.1 [35], and all of the parameters of the descriptors are fine-tuned. The detailed information for the selected descriptors is presented in Table 2.

4.1.2. Evaluation criteria

We use the Precision-Recall Curve (PRC) that is commonly used for the evaluation of local feature descriptors to evaluate the performance of all selected descriptors. The precision is the number of correct matches concerning the total number of matches, and the recall is the number of correct matches with respect to the number of corresponding features. We refer the readers to reference [36] for more details on the generation of PRC.

4.1.3. The experimental procedures

All of the experiments are conducted on an Intel Core i7-4790 3.6 GHz CPU desktop with 8.0GB of RAM, running 64-bit Microsoft Windows 10, Visual Studio 2017 and PCL 1.8.1.

The uniform sampling implemented in PCL is exploited to detect the feature points in each model, and the same parameters are set. The overall performance of the descriptors combined with uniform sampling has been proved to be the lowest compared to those achieved when combined with Harris3D, ISS and ISS-BR [4], due to the low repeatability of the resulting feature points. This is because uniform sampling does not consider the geometric characteristics of the underlying local surface while performing feature

point detection, therefore, both smooth features and sharp features could be extracted. We thus utilize uniform sampling to make a strict evaluation of the performance, by avoiding any performance improvement from the feature point detection process.

The support radius r is also an important parameter for all descriptors, we set support radii for all descriptors on all datasets to 20 d_p to make the test fair and unbiased, and d_p is computed as Eq. (5).

Note that all tested descriptors are computed only for the feature points detected by uniform sampling. Furthermore, all of the models in the tested datasets are randomly rotated and translated, thus, the ground truth correspondences, rotations and translations are known.

Hence, we can consider that the test is fair and unbiased since the same procedure is applied to all descriptors.

4.2. Descriptiveness

In this section, we use PRC to evaluate the descriptiveness of the descriptors on the selected three datasets. The PRC results with the AUCpr shown in parentheses right next to the descriptors are presented in Fig. 4.

4.2.1. Bologna dataset

Fig. 4 (a) gives the PRC results of the selected descriptors on the Bologna Models. We can see FPFH gives the lowest score, since it encodes the least information of the underlying surfaces. All the other descriptors produce very close responses, the PRCs are all closing to the ideal PRC, and the AUCs are all closing to 1. However, the proposed KDDs with different parameters and matching metrics still outperform the SHOT, RoPS and BRoPH. Besides, from Fig. 4(a), two other observations can be made. First, we found that KDD[5]_Cos achieves the best score, closely followed by KDD[5]_Pearson, KDD[5]_ED and KDD[6], meanwhile, they all achieve slightly better results compared to KL divergence, but KDD[5]_KL still outperforms the other descriptors. Second, the results show that the descriptiveness of KDD is improved by a larger *CubeN*, which can be interpreted that larger *CubeN* results in more cubes and more cubes provide more details about the point distribution. This validates the effectiveness of the cubic support region that covers the 3D space around the feature point.

4.2.2. SpaceTime dataset

The SpaceTime dataset was obtained using the SpaceTime Stereo acquisition technique, thus making the point cloud with a

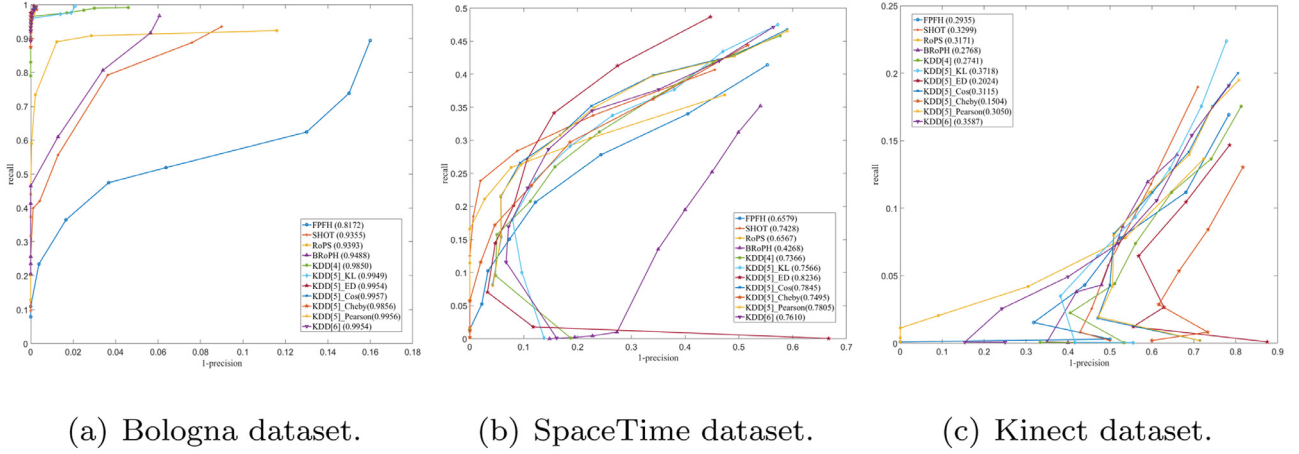


Fig. 4. The descriptiveness results using PRC on the three datasets. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

medium level quality. Fig. 4(b) gives the PRC results of the selected descriptors on the SpaceTime dataset. We can see that the proposed KDD[5]s and KDD[6] all achieve better scores, closely followed by SHOT and KDD[4]. Amongst the KDD[5]s, KDD[5]_ED produces the best score, closely followed by KDD[5]_Cos and KDD[5]_Pearson. Furthermore, FPFH and RoPS produce moderate results, which are better than that of BRoPH. This is contributed to the use of an LRF computed without distance weights on the models with a medium or a lower level sampling qualities (the same conclusion can be made on the Kinect dataset), since the local imbalances in the sampling pattern strongly affect the weights that are computed using distances, and therefore making the LRF more unstable than the one that is computed without distance weights. Besides, note that the recalls achieved by all KDDs are much higher than the other descriptors on this dataset, which clearly demonstrates the effectiveness and advantages of the proposed KDD.

4.2.3. Kinect dataset

Kinect dataset is acquired through a low-cost scanner, i.e. Microsoft Kinect. Thus the quality of the models and scenes is relatively low compared to other two datasets. The PRC results on this dataset are shown in Fig. 4(c). It is found that all descriptors achieve lower recall on this dataset than that on the Bologna dataset and Space Time dataset by a large margin due to the low quality of the models and scenes. However, still the proposed KDD[5]_KL and KDD[6] achieve the best performance. The next best performing descriptors are SHOT and RoPS, closely followed by KDD[5]_Cos and KDD[5]_Pearson. Note that, on this dataset, the KDD[5]_ED and KDD[5]_Cheby give the lowest scores.

Comparing to the results in Fig. 4(a) and (b), the following conclusions can be drawn: (1) difference-based distances (such as Euclidean distance and Chebyshev distance) are not suitable for the point clouds with low resolution qualities, whereas correlation-based (such as Cosin distance and Pearson distance) and entropy-based (for example, KL divergence) distances are more suitable for point clouds with low resolution qualities; (2) larger *CubeN* decreases the score of the KDD on the point clouds with a relatively lower image qualities, this is because more cubes may capture more details about the sampling densities which are not useful information of the underlying surfaces.

4.3. Compactness

Different descriptors have different lengths that affect not only the computational-efficiency and memory footprint but also the

Table 3

The AUCpr results of the descriptors on the three datasets. The 3 best performance is reported in bold face, and the 3 next best performance is reported in *italic*.

Datasets	Bologna	Space Time	Kinect
FPFH	0.8172	0.6579	0.2935
SHOT	0.9355	0.7428	0.3299
RoPS	0.9393	0.6567	<i>0.3171</i>
BRoPH	0.9488	0.4268	0.2768
KDD[4]	0.9850	0.7366	0.2741
KDD[5]_KL	0.9949	0.7566	0.3718
KDD[5]_ED	0.9954	0.8236	0.2024
KDD[5]_Cos	0.9957	0.7845	<i>0.3115</i>
KDD[5]_Cheby	0.9856	0.7495	0.1504
KDD[5]_Pearson	0.9956	0.7805	<i>0.3050</i>
KDD[6]	0.9954	0.7610	0.3587

descriptiveness. However, it's not that the longer the descriptor, the better its performance. The term compactness that is defined as the overall descriptiveness of each floating-point number in a descriptor vector, is also considered an important performance of a descriptor [4]. Specifically, compactness is calculated as the average value of AUCpr in Table 3 divides the length of each descriptor given in Table 2, and the compactness results are shown in Fig. 5.

BRoPH is the most competitive since the length of the binary descriptors is its dimension divides 8, though it gets no best scores on the three selected datasets. The predominant advantage of binary descriptors is their admirable length and the lower memory footprint. FPFH comes second with a very short length, (i.e. the length is 33). The third place is KDD[4], followed by KDD[5]s that achieve close performance in terms of compactness. RoPS and KDD[6] produce moderate scores. Besides, SHOT achieves the lowest compactness compared to all the other descriptors, but it produces relatively high scores concerning the descriptiveness on the Kinect dataset. This is because the dimension of SHOT is enlarged while encoding more information. It can, therefore, be concluded that binary descriptors are quite suitable for applications with strict limits on the storage requirements (e.g., mobile phones), and floating descriptors are more suitable for applications that prefer high precision and recall (e.g., 3D model retrieval and registration).

4.4. Robustness

In this section, we also use PRC to evaluate the robustness of the selected descriptors concerning Gaussian noises and varying sampling densities. To reduce the number of figures and improve

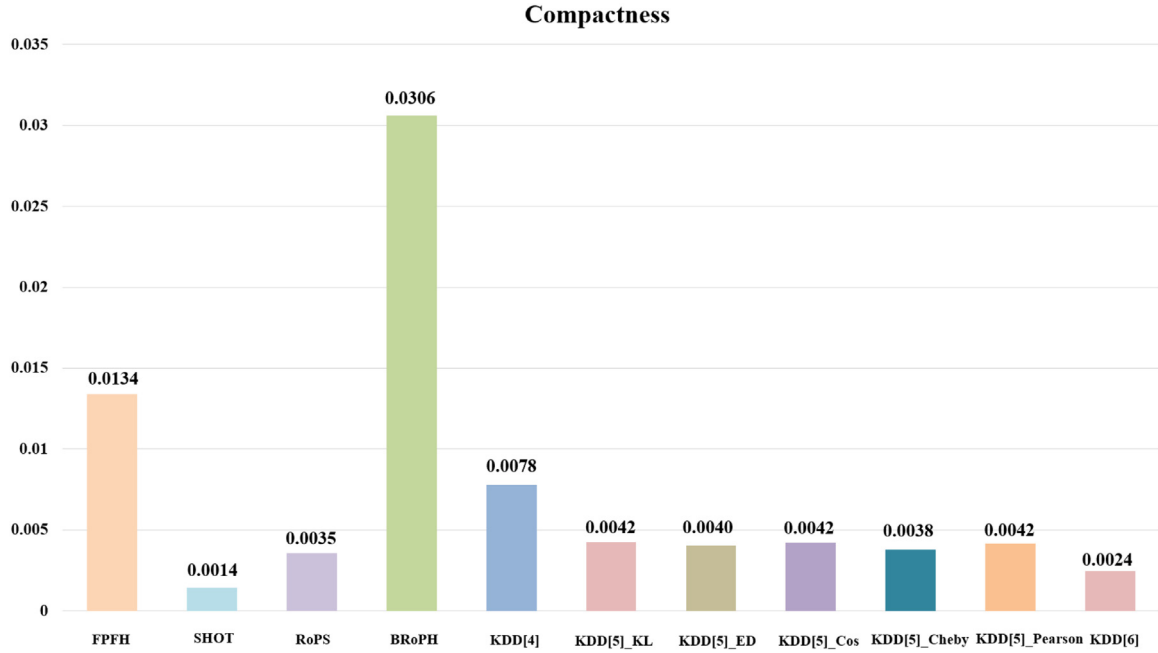


Fig. 5. The compactness results of the selected descriptors on these datasets. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

readability, we only present experimental results on the Space-Time dataset and the Kinect dataset, furthermore, we only test the robustness of KDD[5]_KL since it achieves balanced performance on the selected datasets. The PRC results of the proposed KDD descriptor and the other four compared descriptors on the two datasets are presented in Fig. 6.

4.4.1. The robustness to Gaussian noises

In order to evaluate the robustness to Gaussian noises of the descriptors, we add Gaussian noises with the increasing standard deviation of 0.1, 0.3, and 0.5 d_p to the tested models.

The PRC results at different levels of noises are presented in Fig. 6(a)–(f). It can be found that SHOT and FPFH outperform our KDD at a low level of noise (0.1 d_p) on the Space Time dataset, however, when the noise level increases (i.e. 0.3 d_p and 0.5 d_p), our KDD achieves the best result. Furthermore, regarding aspect of robustness to noises on the Kinect dataset, our KDD perform comparable with respect to low and medium levels of noise, but KDD is inferior to RoPS and SHOT at high noise levels on the Kinect dataset. However, note that, for both of the two selected datasets, our KDD achieves the highest recall at different levels of noises. This can be contributed to the cubic support region and point's density computation of the selected spatial points. In particular, as long as the points move within the cube due to the disturbance on the coordinates, the D_{c_i} will not change.

4.4.2. The robustness to varying sampling densities

The results of Section 4.2 demonstrate the ability to handle the point non-uniformity since the tested models all exhibit strong local imbalances in the sampling pattern and the variation in the image quality of the tested datasets enables us to analyze the performance of the selected descriptors under different levels of image qualities. To further test the robustness against varying sampling densities, we also tested our approach both on the original models and the simplified scenes (at simplification rate of 0.5, 0.7 and 0.9), which present the same surface but with different sampling densities.

The PRC results at diverse simplification rates are presented in Fig. 6(g)–(l). On the SpaceTime dataset, the performance of SHOT and KDD at simplification rate of 0.7 and 0.9 is comparable, which perform better than the other descriptors. On the Kinect dataset, the proposed KDD surpasses the others by a large margin at a high and a medium simplification rate. However, on both two datasets, the feature matching precision of KDD decreases rapidly as the simplification rate further decreases (i.e. at simplification rate of 0.5), which can be considered a drawback of the proposed KDD, though it still achieves the highest recalls.

4.5. Efficiency

The efficiency of the selected descriptors in terms of time spent for feature description and matching is compared.

The generation timings of generating the tested descriptors under different support sizes are presented in Fig. 7. It can be seen that the BRoPH and SHOT are the most efficient among all selected descriptors, followed by RoPS which produces a moderate performance. Behind are FPFH and KDD[4], although their computational performance is close, the KDD[4] is slower than FPFH, especially for local surfaces with 4000 - 8000 points. KDD[5] and KDD[6] become the slowest methods. Furthermore, it can be seen that time spent on generating KDD increases severely as the N increases from 4 to 6, this is because the computation complexity of generating KDD for one feature point is N^3 . Specifically, to build the KDD descriptor for a feature point, the points' densities of $N^3 \times n$ spatial points are computed, and then the kernel densities of N^3 cubes are computed to form the descriptor, thus the time spent on generating KDD significantly depends on N .

The average matching time for the descriptors are listed in Table 4, and their dimensions are listed in Table 2. It is obvious that the length or dimension of a descriptor has a significant impact on matching time. The binary descriptors are undoubtedly more efficient than the floating descriptors in terms of feature matching since they utilize the Hamming distance matching via XOR operation. The time of KDD[4] and KDD[5] spent for matching

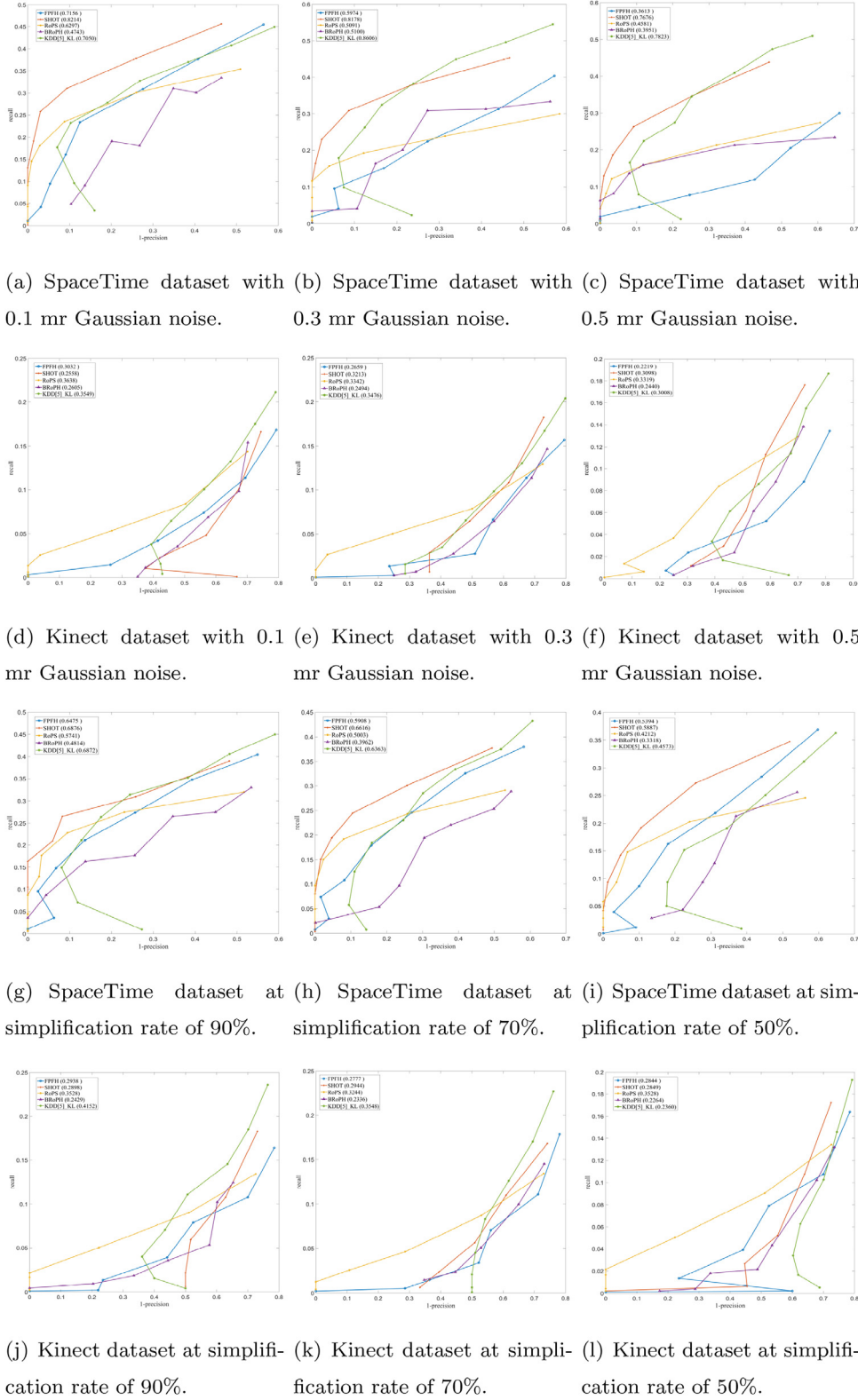


Fig. 6. The robustness testing results concerning Gaussian noises and varying sampling densities. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 4

Average matching time for selected descriptors. The timings are in seconds.

Descriptors	FPFH	SHOT	RoPS	BRoPH	KDD(4)	KDD(5)	KDD(6)
Matching Time	0.091	0.934	0.541	0.036	0.254	0.475	0.668

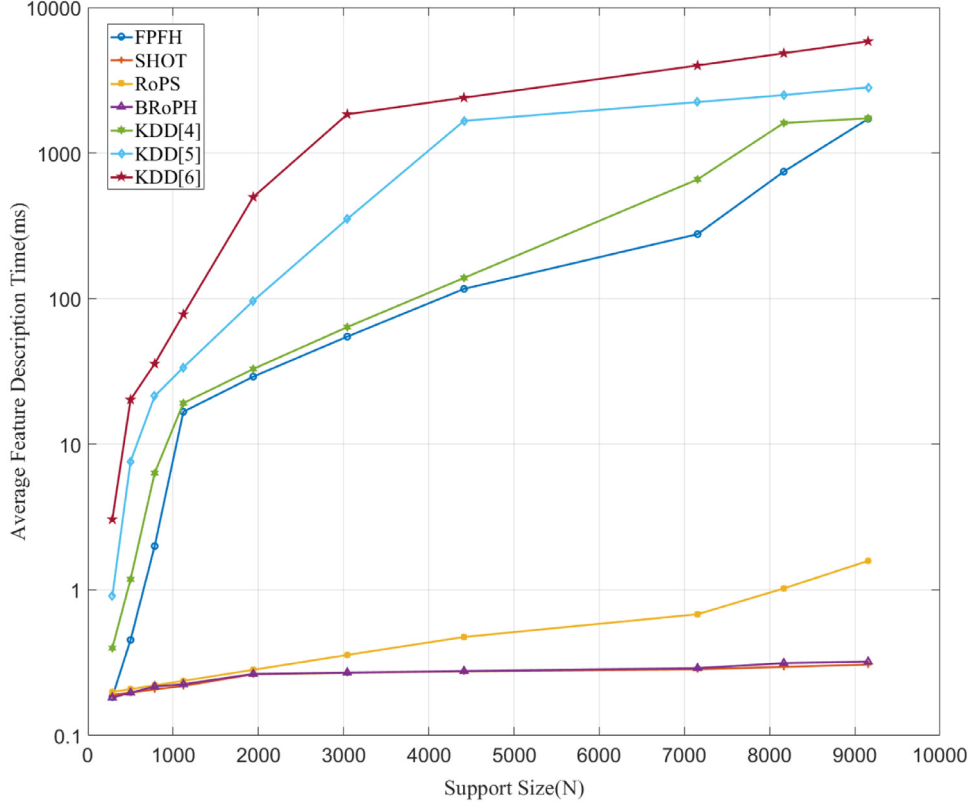


Fig. 7. Computational timings of generating a feature descriptor with varying number of points in the support regions. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

outperforms SHOT and RoPS, furthermore, the time of KDD[6] also outperforms SHOT, but the time of KDD[6] is more than RoPS due to its increasing length.

4.6. Overall performance summary

The KDD is a simple 3D local feature descriptor since it encodes the spatial distribution of points, avoiding computing any geometric attributes and needing no intermediate computations. The KDD is also easily implemented and evaluated by other researchers since its computation only contains two steps.

The proposed descriptor KDD has been designed to capture the information of the whole 3D space by estimating kernel densities of all local cubes (no matter it is empty or not) in the support region, instead of only computing the points' numbers or geometrical information of neighboring points. Thus the KDD descriptor is highly descriptive and the thorough experimental evaluation validates and demonstrates that KDD compares favorably to state-of-the-art 3D description approaches. KDD is also rotation-invariant and robust to Gaussian noises, as well as point non-uniformity and varying sampling densities, though it gets a lower score on sparse point clouds. Furthermore, it also can be seen that the proposed KDD achieves the highest recalls on different datasets, for both at different levels of noises and at different simplification rates, which is a significant advantage.

Moreover, KDD also achieves a high performance concerning the compactness, though the compactness score of KDD is lower than that of BRoPH which is a binary descriptor. The FPFH also outperforms KDD in terms of compactness, but it falls behind KDD in terms of descriptiveness.

The computational-efficiency of KDD is a major drawback since the computation complexity is N^3 , making KDD not suitable for applications with strict limits on the computation speed. However,

for the applications that prefer high recalls, the timing of computing KDD is also acceptable.

Based on the experimental results presented above, three interesting and meaningful points can be summarized. First, the LRF computed without distance weights is more suitable for the model with a medium or a lower level sampling qualities, since the local imbalances in the sampling pattern strongly affect the weights that are computed using distances, and therefore making the LRF more unstable than the one that is computed without distance weights. Second, difference-based, correlation-based and entropy-based distances are suitable for dataset with a high or medium level of resolution quality, and only correlation-based distance (such as Cosin distance and Pearson correlation coefficient) and entropy-based distance (for example, KL divergence) are suitable for dataset with a low level of resolution quality. Third, larger *CubeN* promotes the performance of the KDD on the point clouds with a high resolution quality, whereas larger *CubeN* decreases the score of the KDD on the point clouds with a lower image qualities since more cubes capture information of the point non-uniformity which is not useful information.

5. Applications

The output of our method is a set of point-to-point correspondences on two given point clouds. A lot of techniques that rely on point sets apply the point-to-point correspondences as input data, such as 3D object registration [3], 3D object recognition [2] and 3D object reconstruction [37]. In this section, some matching results on the public datasets and real-world datasets are presented to intuitively demonstrate the effectiveness of KDD and highlight the utility of the proposed method.

Given two point clouds, we first detect the feature points using uniform-sampling, then the feature points are described with KDDs

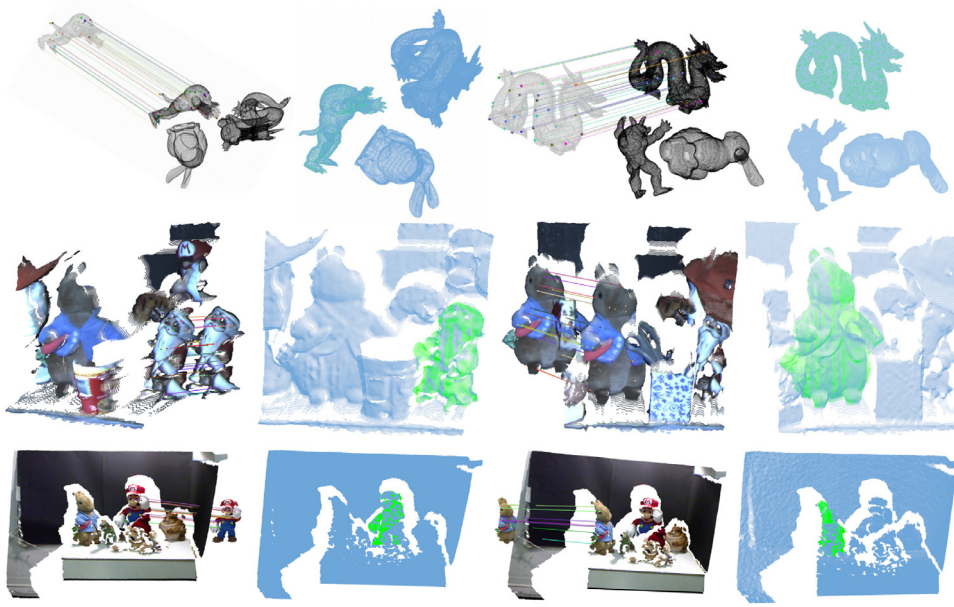


Fig. 8. 3D object registration and recognition results. From top to bottom row are the results on Bologna, SpaceTime and Kinect datasets respectively. From left to right in each row, point-to-point correspondences, coarse registration result, another pair of correspondences and coarse registration result. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

and further matched by computing the KL divergence of the KDDs. The Random sample consensus (RANSAC) implemented in PCL is utilized to detect the inlier correspondences that used to estimate a coarse transformation between the two given point clouds.

5.1. 3D object registration and recognition

By applying the above procedures on the public datasets that are exploited in Section 4, 3D object registration and recognition results are obtained in Fig. 8. It can be seen from the figure that some corresponding point pairs have been detected using the proposed KDD and the corresponding point pairs are good enough to trigger RANSAC to get a coarse transformation estimation. Consequently, source point clouds are aligned with the target point clouds correspondingly, which can serve to finely align point clouds and recognize 3D objects.

5.2. 3D object reconstruction

Point density variation proves to be a major cause of failure for 3D descriptors. However, it occurs quite often with real data, due to uneven scanning proficiency of scanning personnel and the scanning techniques employed by the scanning devices. In order to further confirm the effectiveness of the proposed KDD, we apply it to the challenging real-world datasets, i.e. Terracotta fragment models, which were scanned by students using a Creaform VIU handy scanner. The scan resolution was 3.91 mm, which favors speed but consequently results in quite rough precision.

In general, 3D laser scanners can only obtain the point cloud data of a 3D object in a certain view, therefore, point clouds from multiple views need to be integrated to create a complete 3D model of an object. Point clouds obtained from multiple views often share common surfaces, thus the point-to-point correspondences on the sharing surface can be applied to get a coarse transformation estimation. The complete 3D model thereby is obtained by aligning the point clouds obtained from multiple views. Following the above procedures, we apply the KDD to Terracotta fragment models to get a coarse transformation estimation between

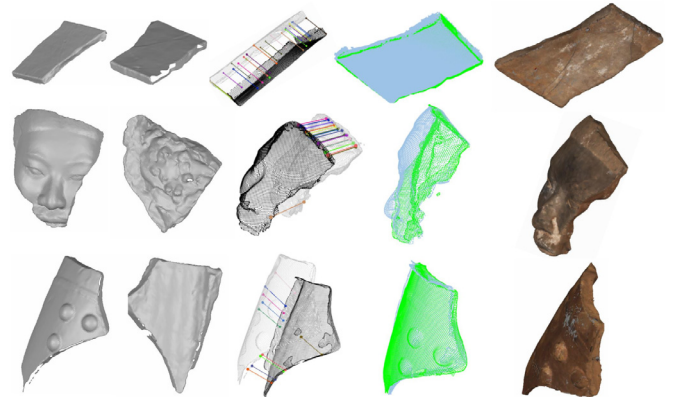


Fig. 9. 3D object reconstruction results. From top to bottom rows are three different Terracotta fragments. From left to right in each row, two pieces of scanning data, point-to-point correspondences, coarse registration result and complete fragments with texture. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

two pieces of point clouds that are obtained from two different views. The point clouds exhibit strong local imbalances in the sampling pattern because they were scanned by the students who are not proficient in scanning. Besides, the point clouds contain realistic noises, which were brought during the scanner acquisition process. However, here again, the proposed method delivers quite favorable results as shown in Fig. 9. It can be observed that the KDD consistently provides a descriptive and robust description of the feature points in the presence of realistic noises and strong local imbalances in the sampling pattern. Hence the effectiveness of the proposed method is solidly validated.

6. Conclusion

We presented a 3D local feature descriptor that is termed KDD. In our method, we first compute the LRFs at key points and build the cubic support region that serves to the kernel density estimation and meanwhile promote the robustness of the proposed

KDD descriptor. Based on the cubic support region, we estimate the kernel densities of the local cubes. Then the density estimates are aggregated into a vector and thereby forming the KDD descriptor. Consequently, the true point correspondences can be achieved via KL divergence. The KDD descriptor has proven to be highly descriptive, furthermore, KDD achieves the highest recalls compared to the tested descriptors on the selected datasets, for both at different levels of noises and at different simplification rates. Besides, the KDD provides invariance to rigid transformations and achieves satisfactory results in terms of robustness and compactness. However, the computational efficiency of KDD is considered a drawback, thus as for future work, we would like to speed up the generation of KDD. Furthermore, one very interesting future work direction is applying the kernel density estimation with a cubic support region to the distribution-preserving sampling and generation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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